Trainings

New Engine Break-in Oils

⚠ CAUTION ⚠

A sulfated ash limit of 1.85 percent has been placed on all engine lubricating oils recommended for use in Cummins® engines. Higher ash oils can cause valve and/or piston damage and lead to excessive oil consumption.

⚠ CAUTION ⚠

The use of a synthetic-based oil does not justify extended oil change intervals. Extended oil change intervals can decrease engine life due to factors such as corrosion, deposits, and wear.

Special break-in engine lubricating oils are **not** recommended for new or rebuilt Cummins® engines. Use the same type of oil during the break-in as used in normal operation.

Additional information regarding lubricating oil availability throughout the world is available in the Lubricating Oils Data Book for Heavy-Duty Automotive and Industrial Engines. It can be ordered from: Engine Manufacturers Association (EMA), Two North LaSalle Street, Chicago, IL 60602; (www.engine-manufacturers.org)

Precautions and Instructions for Proper Kit Use

If an engine is operated in ambient temperatures consistently below -23°C [-9°F], and there are no provisions to keep the engine warm when it is **not** in operation, use a synthetic CE/SF or higher American Protroleum Institute (API) classification engine oil with adequate low-temperature properties such as 5W-20 or 5W-30.

The oil supplier is responsible for meeting the performance service specification represented with its product.

General Information

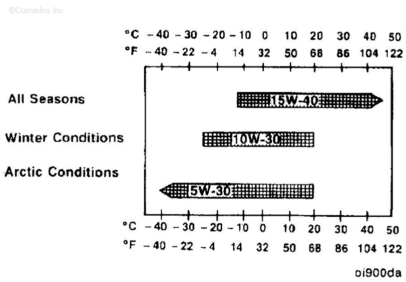
The use of quality engine lubricating oils, combined with appropriate oil drain and filter change intervals, are critical factors in maintaining engine performance and durability.

Cummins® recommends the use of a high-quality SAE 15W-40 multiviscosity heavy-duty engine oil, such as Cummins Premium Blue®, that meets the requirements of Cummins® Engineering Specification CES20071 or CES20076, or the American Petroleum Institute (API) performance classification CG-4 or CH-4.

A sulfated ash limit of 1.0 mass percent is suggested for optimum valve and piston deposit and oil consumption control. The sulfated ash **must not** exceed 1.85 mass percent.

For further details and discussion of engine lubricating oils for Cummins® engines, refer to Cummins® Engine Oil Recommendations, Bulletin Number 3810340, or a Cummins® Authorized Repair Location.

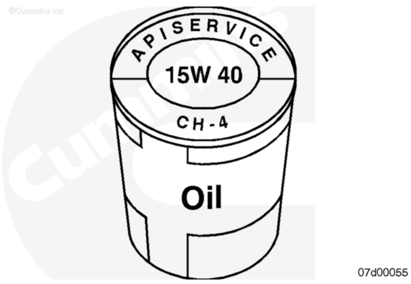
The use of low-viscosity oils, such as 10W or 10W-30, can be used to aid in starting the engine and in providing sufficient oil flow at ambient temperatures below -5°C [23°F]. However, continuous use of low-viscosity oils can decrease engine life due to wear. Reference the accompanying chart.



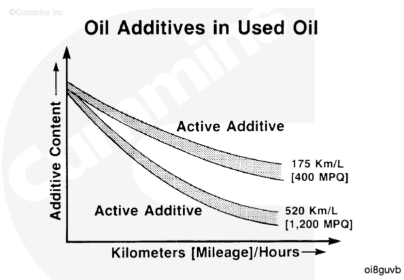
The API service symbols are shown in the accompanying illustration. The upper half of the symbol displays the appropriate oil categories.

The lower half can contain a description of oil energy conserving features.

The center section identifies the SAE oil viscosity grade.



As the engine oil becomes contaminated, essential oil additives are depleted. Lubricating oils protect the engine as long as these additives are functioning properly. Progressive contamination between oil and filter change intervals is normal. The amount of contamination will vary depending on the operation of the engine, kilometers or [miles] on the oil, fuel consumed, and new oil added.



Extending oil and filter change intervals beyond the recommendations will decrease engine life due to factors such as corrosion, deposits, and wear.

Use the following procedure to determine which oil drain interval to use for an application.

For C8.3 Industrial:

See the C8.3 Industrial and Generator Drive Operation and Maintenance Manual, 2883407. Refer to Procedure 018-003 in Section V. (/qs3/pubsys2/xml/en/procedures/41/41-018-003.html)

For C8.3 Commercial Marine:

See the C8.3 Commercial Marine and Industrial Operation and Maintenance Manual, 4021330. Refer to Procedure 018-003 in Section V. (/qs3/pubsys2/xml/en/procedures/41/41-018-003.html)

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